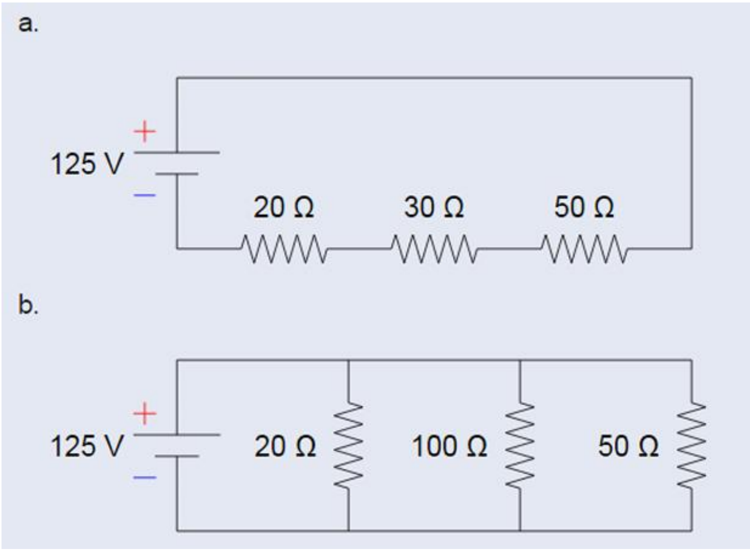
ICS 4111: Embedded Systems & IoT

Practical Exercise 2: Circuit Design

Instructions:

For this exercise, you will largely use the EasyEDA software for circuit design. In part A, you can write the results of your calculations to a suitable document and export it as a PDF. For part B, export the designs directly from EasyEDA as PDF documents.

Part A



For each of these circuits:

1. Calculate the total resistance in ohms,
2. Calculate the total current from the power supply,
3. Calculate the current through each resistor
4. Calculate the Voltage drop through each resistor
5. Calculate the Power dissipated through each resistor *(use Watt's Law: Power = Voltage * Current*)

Part B

1. Develop schematic diagrams that describe the 2 circuits in Part A and include tables that list all components per schematic. See the example given here

2. Design a schematic diagram for an Arduino Nano microcontroller connected to a DHT22 temperature and humidity sensor. For this schematic as well, develop a table that lists all the components. Find a reference schematic for the Arduino Nano here. (*Note: DHT22 requires 3.3v-5v power supply*)

Once done, publish your work to your GitHub account by following these steps:

1. Access your GitHub account and create a new repository: **Circuit-Design**
2. Upload all your documents to the repo.
3. After confirming the work is on your GitHub repo, copy the link to the repo from your address bar as *github.com/AllanVikiru/Circuit-Design* and submit this link on e-learning before the due date. (confirm the appropriate due dates for your group on e-learning).

Marking points for this work are correctness and completeness of the calculations and circuits.

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